

Erdős Problem #679

EP-679

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\epsilon > 0$ and $\omega(n)$ count the number of distinct prime factors of n . Are there infinitely many values of n such that $\omega(n - k) < (1 + \epsilon) \frac{\log k}{\log \log k}$ for all $k < n$ which are sufficiently large depending on ϵ only?

Can one show the stronger version with $\omega(n - k) < \frac{\log k}{\log \log k} + O(1)$ is false?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>